MODULE 45 · EXAM STRATEGY - ALL SECTIONS

Practice Paper 3

The final twenty new questions, across all five sections, then a way to read your three scores together and decide what to study next.

SECTIONS All five MODULE 45 of 45 QUESTIONS 20

admin@sthithapragna.com

How to use this paper

- Nine seconds per question. **Pause the video** if you need longer.
- Read the **ask**, find the **qualifier**, strike what breaks a fixed fact.
- One question pairs with paper 2. Spot the **constraint** that changed.
- Keep your scores from papers 1 and 2 to hand for the **final slide**.

> REMEMBER
>
> Treat this paper like the real exam: no notes, and no rewinding the question.

> Spark jobs from the data science team write Larkspur's settlement tables to Cloud Storage. Analysts must query the same tables in BigQuery.
>
> The tables must stay in an open format, with one copy and one catalog.

Which design fits best?

A Parquet files in Cloud Storage, plus separate copies loaded into BigQuery

B BigQuery native tables, exported to Parquet each night for Spark

C Hive tables on a long-running Spark cluster, queried by federation

D Iceberg tables in the Lakehouse runtime catalog, read by both engines

◆ PAUSE AND DECIDE

Which design fits best?

A Two copies

Two copies to keep in step.

B Nightly export

A copy, a day stale for Spark.

C Hive on a cluster

A cluster to keep running for the catalog.

D Iceberg and the runtime catalog

One open format, one catalog, two engines.

THE SIGNAL

One open copy for Spark and BigQuery: Iceberg in the Lakehouse runtime catalog.

> The app team wants to measure how long customers spend in the Larkspur app per visit. A visit ends after 20 minutes with no activity.
>
> Click events stream in through Pub/Sub.

Which windowing should the Dataflow pipeline use?

A Fixed 20-minute windows, keyed by customer and app screen

B Session windows keyed by customer, with a 20-minute gap

C Sliding windows of 20 minutes, starting every minute

D A single global window, triggered every 20 minutes

◆ PAUSE AND DECIDE

Which windowing should the Dataflow pipeline use?

A Fixed windows
Clock slices cut visits in half.

B Session windows
One window per visit, closed by inactivity.

C Sliding windows
Overlapping slices, not visits.

D Global window
One window for all time.

THE SIGNAL

A visit that ends after inactivity: session windows with a gap.

A new Spanner table records meter readings at a very high write rate.

The first design uses an auto-incrementing reading number as the primary key, and writes queue on one server.

What should the team change?

A Use the reading timestamp as the primary key instead

B Keep the sequence, but sort the key in descending order

C Use a version 4 UUID as the primary key

D Add more nodes, so the hot split gets more capacity

◆ PAUSE AND DECIDE

What should the team change?

A Timestamp key

Still increasing, still one hot split.

B Descending sequence

Descending is still monotonic.

C UUID version 4

Random, so writes spread across splits.

D More nodes

A hotspot stays on one split.

THE SIGNAL

Monotonic keys make hotspots: a random UUID, or a bit-reversed sequence.

> Finance analysts work in Google Sheets. They want pivot tables over a 2-billion-row BigQuery bills table.
>
> They do not write SQL, and nothing should be exported.

What should they use?

A Connected Sheets, with pivots computed in BigQuery

B Export the table to CSV, and import it into Sheets

C A scheduled query writing 2 billion rows into a sheet

D Data Studio, embedded as an image in each sheet

◆ PAUSE AND DECIDE

What should they use?

A Connected Sheets
Sheets interface, BigQuery does the work.

B Export to CSV
Sheets cannot hold billions of rows.

C Scheduled query
Far beyond any sheet's limit.

D Embedded image
A picture, not a pivot.

THE SIGNAL

Sheets users, warehouse-sized data, no SQL: Connected Sheets.

> On one Enterprise reservation, heavy nightly ETL now overruns into the morning, and executive dashboards slow down until it finishes.
>
> Dashboards must stay fast, without starving ETL of capacity at night.

What should the team do?

A Move every dashboard project to on-demand pricing

B Separate etl and bi reservations, each with its assignments

C Raise the reservation's maximum, and hope the queries share well

D Run all the ETL jobs as batch priority queries

◆ PAUSE AND DECIDE

What should the team do?

A On-demand

Gives up predictable cost for one clash.

B Two reservations

Each workload has its own slots, idle ones shared.

C Raise the maximum

More cost, the same contention.

D Batch priority

Delays ETL, but guarantees nothing.

THE SIGNAL

Workloads competing for slots: separate reservations and assignments.

> Smart thermostats publish JSON events to Pub/Sub. The message schema already matches a BigQuery table, and no transformation is needed.
>
> The team wants the events in BigQuery, with nothing to run or maintain.

What should the team set up?

A. A BigQuery subscription that writes into the table

B. A Dataflow streaming job from a Google-provided template

C. A Cloud Run service that inserts each event through the API

D. A Pub/Sub Lite reservation feeding BigQuery

◆ PAUSE AND DECIDE

What should the team set up?

A BigQuery subscription
Pub/Sub writes directly, no subscriber to run.

B Dataflow template
Works, but a pipeline to run and pay for.

C Cloud Run service
Code to write and scale.

D Pub/Sub Lite
Turned down in March 2026.

THE SIGNAL

No transformation, nothing to run: a BigQuery subscription.

> Larkspur is retiring an on-premises Teradata warehouse with 4,000 SQL scripts written in Teradata's dialect.
>
> The scripts must run on BigQuery, and the team wants to avoid rewriting them by hand.

What should the team use?

A. Database Migration Service, pointed at the Teradata system itself

B. Datastream, replicating Teradata into BigQuery

C. A team of contractors rewriting each script by hand

D. BigQuery Migration Service and its batch SQL translator

◆ PAUSE AND DECIDE

What should the team use?

A DMS

Moves operational databases to Cloud SQL or AlloyDB.

B Datastream

Not a Teradata source, and moves data, not SQL.

C Manual rewrite

Slow, costly and error-prone.

D Batch SQL translator

Translates the dialect in bulk.

THE SIGNAL

Warehouse SQL in another dialect: BigQuery Migration Service translation.

> Larkspur's readings dataset holds 800 TB of logical bytes, but the data compresses about ten to one. Tables rarely change after loading.
>
> The team wants to cut storage cost without changing any queries.

What should the team change?

A Move the tables to Cloud Storage, as compressed CSV files

B Shorten the time travel window, on the logical billing model

C Switch the dataset to the physical storage billing model

D Delete partitions older than 90 days to save space

◆ PAUSE AND DECIDE

What should the team change?

A CSV in Cloud Storage

Changes every query, and loses native tables.

B Shorter time travel

Under logical billing, it is already included.

C Physical billing

Billed on compressed bytes, roughly a tenth.

D Delete partitions

Deletes data still needed.

THE SIGNAL

Highly compressible data: the physical storage billing model.

> Several curated tables feed regulatory reports. The team needs rules, such as non-null meter IDs and kWh within a range, checked on a schedule.
>
> Results must be recorded, and failures must raise alerts.

What should the team use?

A Knowledge Catalog data quality scans with those rules

B Analysts spot-checking a sample of each report every Monday

C Raising max_bad_records on the load jobs

D Pub/Sub schemas on the topics feeding the tables

◆ PAUSE AND DECIDE

What should the team use?

A Data quality scans

Rules, schedules, scorecards and alerts, managed.

B Spot checks

Manual, and misses most errors.

C max_bad_records

Drops bad rows instead of reporting them.

D Pub/Sub schemas

Checks message shape, not values in tables.

THE SIGNAL

Scheduled rules on tables, with results and alerts: data quality scans.

> A custom settlement job writes the line SETTLEMENT_MISMATCH to Cloud Logging whenever totals disagree.
>
> On-call staff must be paged if it appears more than five times in ten minutes.

What should the team set up?

A. A daily export of logs to BigQuery, checked each morning

B. An uptime check on the settlement job's endpoint

C. A log-based metric for that line, plus an alert

D. A budget alert on the settlement project

◆ PAUSE AND DECIDE

What should the team set up?

A Daily export

A day too late to page anyone.

B Uptime check

Checks the job is up, not what it logs.

C Log-based metric

Counts matching lines, alerts on the threshold.

D Budget alert

Spending, not errors.

THE SIGNAL

Page on a log line pattern: a log-based metric and an alerting policy.

> Analysts must join customer records across three tables on email address, but must never see real email addresses.
>
> The support team must still see emails in full.

What should the team set up?

A Policy tags on email, with no access granted to analysts

B Dynamic data masking with the SHA-256 rule for analysts

C Dynamic data masking with the nullify rule for analysts

D A Sensitive Data Protection job that redacts email in place

◆ PAUSE AND DECIDE

What should the team set up?

A Policy tags only

Queries on email fail, so no joins.

B SHA-256 masking

Same email, same hash, so joins still work.

C Nullify masking

Every email becomes null, so joins break.

D Redact in place

Support loses the real values too.

THE SIGNAL

Hidden but joinable: dynamic masking with SHA-256.

Security found a BigQuery dataset shared with a personal Gmail address.

Larkspur wants to make it impossible to grant any access to identities outside its own domain, across the whole organisation.

What should the team do?

A Review every dataset's permissions once a month

B Remove the personal account, and email staff a reminder

C Put the dataset inside a VPC Service Controls perimeter

D Set the domain restricted sharing organisation policy

◆ PAUSE AND DECIDE

What should the team do?

A Monthly review

Finds problems after the fact.

B Remove and remind

Fixes one case, prevents nothing.

C Perimeter

Blocks data movement, not IAM grants.

D Domain restricted sharing

Grants to outside identities are refused.

THE SIGNAL

No access for outside identities, ever: domain restricted sharing.

> Yesterday, an engineer deleted a folder of settlement files from a Cloud Storage bucket by mistake.
>
> The bucket was created recently with default settings, and object versioning is off.

How can the team recover the files?

A They cannot, because without versioning deleted objects are gone

B Ask Customer Care to restore them from fail-safe

C Restore them from soft delete, within the retention window

D Use BigQuery time travel on the bucket

◆ PAUSE AND DECIDE

How can the team recover the files?

A Gone for good

Soft delete keeps them, even without versioning.

B Fail-safe

A BigQuery feature, not Cloud Storage.

C Soft delete

Deleted objects restorable for 7 days by default.

D Time travel

For BigQuery tables, not buckets.

THE SIGNAL

Recently deleted objects in a new bucket: restore from soft delete.

> A nightly join between bills and accounts suddenly takes four times longer.
>
> In the execution graph, one join stage shows a maximum compute time far above its average.

What is the most likely cause?

A. The query cache has been disabled for the project

B. The reservation has too few slots for any query

C. A missing partition filter on the accounts table

D. Data skew: a few join keys carry most of the rows

◆ PAUSE AND DECIDE

What is the most likely cause?

A Cache disabled

Affects repeats, not one stage's balance.

B Too few slots

Slows every stage evenly.

C Missing filter

Shows as more bytes read, not one skewed stage.

D Data skew

A few workers do most of the work.

THE SIGNAL

Maximum far above average in one stage: data skew.

> Larkspur wants an alert message published whenever a newly arriving reading row in BigQuery shows over 50 kWh in a half hour.
>
> Analysts want to write the logic in SQL. The project already runs on an Enterprise reservation.

What should the team build?

A. A continuous query that writes matching rows to Pub/Sub

B. A scheduled query every hour that emails a CSV file

C. An on-demand query, run by hand each morning

D. A Dataflow pipeline, with the SQL logic rewritten in Java

◆ PAUSE AND DECIDE

What should the team build?

A Continuous query
SQL, running continuously, output to Pub/Sub.

B Hourly query
Up to an hour late.

C Run by hand
Manual, and a day late.

D Dataflow in Java
Works, but not SQL.

THE SIGNAL

Act on new rows as they arrive, in SQL: a continuous query.

> Developers need a writable copy of the 5 TB production bills table, to test a schema change. They will modify only a few thousand rows, and storage cost should stay low.

What should they create?

A A full table copy, into a development dataset

B A table clone, in a development dataset

C A table snapshot, then run the tests on it

D A view over the production table

◆ PAUSE AND DECIDE

What should they create?

A Full copy

Pays for all 5 TB again.

B Table clone

Writable, billed only for what changes.

C Table snapshot

Read-only, so tests cannot write.

D View

Reads production, and cannot be written.

THE SIGNAL

Writable test copy, low cost: a table clone.

Larkspur receives 30,000 free-text complaints a month, stored in BigQuery.

The team wants each one labelled billing, metering, supply or service, using SQL, without training a model.

What should the team use?

A A k-means model on the complaint text, in BigQuery ML

B A logistic regression model, trained on labelled examples

C A Gemini model called from SQL with AI.GENERATE_TEXT

D Regular expressions searching each complaint for keywords

◆ PAUSE AND DECIDE

What should the team use?

A k-means
Clusters, but not your four labels.

B Logistic regression
Needs labelled training data.

C Gemini from SQL
A prompt with the four labels, no training.

D Regular expressions
Brittle, and misses paraphrases.

THE SIGNAL

Label free text into set categories, no training: a Gemini model from SQL.

> A Dataform table rebuilds three years of readings every hour, although only the last hour's rows are new.
>
> Costs and run times keep rising as the data grows.

What should the team change?

A Make it an incremental table, adding only new rows each run

B Keep the full rebuild, but run it once a day instead of hourly

C Move the transformation into a Python script on a VM

D Turn on the query cache for the Dataform project

◆ PAUSE AND DECIDE

What should the team change?

A Incremental table
Processes only the new hour.

B Daily rebuild
Cheaper, but up to a day stale.

C Python on a VM
Leaves BigQuery, and adds a server.

D Query cache
Changed data defeats the cache.

THE SIGNAL

Only new rows each run: a Dataform incremental table.

> Larkspur runs a long-lived Spark cluster sized for the nightly peak, with its data in HDFS on the cluster's disks.
>
> The cluster sits idle 20 hours a day, and the team wants to cut cost.

What should the team change?

A. Keep the cluster, and switch every worker to a larger machine type

B. Keep the cluster, but turn off autoscaling to keep costs fixed

C. Move the data to Bigtable, and keep the cluster for compute

D. Keep data in Cloud Storage, and create a cluster per nightly run

◆ PAUSE AND DECIDE

What should the team change?

A Larger machines

A bigger idle cluster.

B No autoscaling

Fixed cost, still idle.

C Bigtable

The wrong store for Spark files.

D Cluster per run

Pay only while jobs run. Data outlives clusters.

THE SIGNAL

Idle most of the day: data in Cloud Storage, clusters per job.

> Larkspur needs dev, test and prod environments with identical datasets, buckets, topics and IAM.
>
> Every change must be reviewed before it reaches prod.

What should the team use?

A Click the resources together in the console for each environment

B Terraform in Git, applied per environment through a pipeline

C A shell script of gcloud commands, run by whoever is on call

D Copy the prod project once, then edit each copy by hand

◆ PAUSE AND DECIDE

What should the team use?

A Console clicks

Environments drift, and nothing is reviewed.

B Terraform in Git

Same code, reviewed, applied per environment.

C Shell script

Not declarative, and depends on who runs it.

D Hand edits

Copies drift at once.

THE SIGNAL

Identical, reviewed environments: infrastructure as code.

Paper 3 answers

QUESTIONS	ANSWERS
1 to 5	D, B, C, A, B
6 to 10	A, D, C, A, C
11 to 15	B, D, C, D, A
16 to 20	B, C, A, D, B

> **REMEMBER**
>
> Note your total and your section scores. The next slides read all three papers together.

Where your marks came from

SECTION	QUESTIONS	REVISIT
1 Design, 22%	1, 7, 12, 19	Modules 02 to 09
2 Ingest and process, 25%	2, 6, 9, 15, 18	Modules 10 to 18
3 Store, 20%	3, 8, 13, 16	Modules 19 to 26
4 Analyse, 15%	4, 11, 17	Modules 27 to 32
5 Maintain and automate, 18%	5, 10, 14, 20	Modules 33 to 39

THE NUMBERS

Add each section across all three papers. Sixty questions give a far steadier picture than twenty.

Reading your three scores together

IF YOU SEE	THEN
One section low in two or three papers	That is your gap. Rewatch its modules, then retake.
Misses scattered across every section	Slow down. Find the qualifier before reading options.
Paired questions answered the same way	You are matching settings. Rewatch module 42.
Scores rising from paper 1 to paper 3	It is sticking. Keep revising the weakest section.
17 or more on every paper	Strong. Plan your exam date, and keep sharp.

HONEST LIMIT

Google publishes no pass mark. These papers measure readiness for study, not a guaranteed result.

Module 45 in five lines

- One open copy for Spark and BigQuery: **Iceberg** in the Lakehouse runtime catalog.
- New rows acted on in SQL: **continuous queries**. No transformation: a **BigQuery subscription**.
- Hidden but joinable: **SHA-256 masking**. Deleted objects: **soft delete**.
- Writable cheap test copy: a **clone**. One stage far above average: **skew**.
- Read your **three scores together**, fix the weakest section, then book the exam.

> COMING UP
>
> That completes the forty-five modules. Module 00 introduces the course and is built last.